4-DP-Longest non-decreasing Subsequence

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer:(penalty regime: 0 %)

Feedback

```
#include <stdio.h>


int main()
{
   int n;
   scanf("%d", &n);
```

```
int a[n], dp[n];

for (int i = 0; i < n; i++)
{
   scanf("%d", &a[i]);
   dp[i] = 1;
}

int max = 1;

for (int i = 1; i < n; i++)
{
   for (int j = 0; j < i; j++)
   {
      // <= because it is NON-DECREASING
      if (a[j] <= a[i])
      {
         if (dp[j] + 1 > dp[i])
            dp[i] = dp[j] + 1;
      }
   }

   if (dp[i] > max)
      max = dp[i];
}

printf("%d", max);
```

```
    return 0;
}
```

	Input	Expected	Got	
	9 -1 3 4 5 2 2 2 2 3	6	6	
	7 1 2 2 4 5 7 6	6	6	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Competitive Programming